

COLMAR-MEYENHEIM AB, France

WMO: 071970

Lat: 47.9286N

Lon: 7.4078E

Elev: 211

StdP: 98.81

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -8.4	(c) -6.2	(d) -10.7	(e) 1.5	(f) -7.0	(g) -8.5	(h) 1.9	(i) -4.3	(j) 13.0	(k) 9.2	(l) 11.5	(m) 8.1	(n) 1.9	(o) 20	(p) 0.517

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.9	32.6	21.0	30.2	20.2	28.5	19.7	22.0	30.1	21.1	28.3	20.4	26.9	3.3	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.5	14.6	24.7	18.7	13.9	23.7	18.0	13.3	23.0	65.0	30.2	62.0	28.3	59.3	27.0	25.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 10.2	(b) 8.9	(c) 7.7	(e) -11.5	(f) 35.1	(g) 2.5	(h) 2.2	(i) -13.3	(j) 36.7	(k) -14.8	(l) 38.0	(m) -16.1	(n) 39.2	(o) -17.9	(p) 40.8		
(a) 10.2	(b) 8.9	(c) 7.7	(e) -11.9	(f) 23.6	(g) 2.4	(h) 0.9	(i) -13.6	(j) 24.2	(k) -15.1	(l) 24.7	(m) -16.4	(n) 25.2	(o) -18.2	(p) 25.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.3	2.2	3.7	7.0	10.7	15.3	18.7	20.4	20.0	15.6	11.5	6.1	3.3		
	DBStd	7.51	4.81	4.20	3.89	3.74	3.52	3.59	3.13	3.26	3.26	3.76	4.11	4.34		
	HDD10.0	937	244	179	107	36	3	0	0	0	1	28	128	211		
	HDD18.3	2809	501	409	350	228	105	39	15	19	94	212	368	467		
	CDD10.0	1397	1	3	15	58	169	261	322	310	169	76	10	3		
	CDD18.3	226	0	0	0	1	13	51	78	71	12	1	0	0		
	CDH23.3	2454	0	0	0	29	183	544	812	726	146	15	0	0		
	CDH26.7	849	0	0	0	2	44	182	302	283	34	1	0	0		
(14) Wind	WSAvg	3.2	3.7	3.8	3.9	3.4	3.2	2.8	2.8	2.5	2.7	2.9	3.3	3.6		
(15) Precipitation	PrecAvg	716	44	42	42	51	78	71	72	70	62	69	57	58		
	PrecMax	944	116	107	162	120	155	144	158	158	147	195	145	128		
	PrecMin	472	10	3	7	8	30	30	29	13	16	2	0	0		
	PrecStd	121	29	25	34	31	35	30	30	33	30	36	34	35		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.8	17.4	21.8	26.5	30.3	33.8	35.6	35.7	30.1	25.6	18.5	14.1		
		MCWB	10.0	10.1	12.4	15.4	19.2	21.7	21.6	21.4	18.7	16.9	13.2	10.7		
	2%	DB	12.1	14.4	18.6	23.4	27.6	31.0	32.4	32.3	27.1	22.1	15.6	12.5		
		MCWB	9.5	9.1	11.2	13.9	17.8	20.3	21.1	20.9	18.9	16.0	11.9	10.0		
	5%	DB	10.5	12.2	15.8	20.7	25.3	28.7	29.7	29.9	24.8	19.8	13.6	11.1		
		MCWB	8.2	8.5	10.1	12.7	16.8	19.8	20.3	20.3	17.6	15.0	10.9	9.1		
	10%	DB	8.7	10.2	13.6	18.4	22.9	26.6	27.9	27.6	22.4	17.6	11.9	9.6		
		MCWB	6.8	7.5	9.2	11.9	15.6	19.0	19.8	19.7	16.4	14.0	9.8	8.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.3	11.0	13.4	16.2	20.4	23.2	23.1	22.7	20.7	18.1	14.3	11.7		
		MCDB	12.8	15.0	19.4	24.9	28.0	31.2	31.8	31.0	26.9	23.3	17.1	13.3		
	2%	WB	9.8	9.9	12.0	14.6	18.9	21.4	22.0	21.7	19.2	16.6	12.6	10.5		
		MCDB	11.8	13.5	16.4	21.5	26.1	28.7	30.1	29.9	25.4	20.5	14.9	12.1		
	5%	WB	8.3	8.8	10.9	13.4	17.5	20.5	21.1	20.9	18.2	15.6	11.2	9.3		
		MCDB	10.2	12.0	14.7	19.7	23.4	27.0	28.3	28.0	23.4	18.9	13.1	10.9		
	10%	WB	7.0	7.6	9.7	12.3	16.3	19.4	20.2	20.1	17.1	14.4	10.0	8.1		
		MCDB	8.5	10.1	13.3	17.4	21.4	25.4	26.6	26.4	21.2	17.2	11.8	9.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.2	8.2	10.1	11.7	11.2	11.6	11.9	12.1	11.2	9.7	7.2	5.9		
		MCDBR	7.8	12.1	14.6	16.6	15.7	15.7	15.8	16.4	15.4	13.3	9.7	7.4		
		MCWBR	5.7	7.7	8.4	8.3	7.2	6.8	6.3	6.4	7.4	7.6	6.5	5.4		
		MCWBR	7.3	10.6	12.1	14.7	13.9	14.3	14.3	14.6	13.1	11.8	9.0	6.6		
(40) Clear-Sky Solar Irradiance	taub	taub	0.314	0.338	0.370	0.406	0.415	0.421	0.418	0.413	0.389	0.367	0.334	0.310		
		taud	2.450	2.380	2.315	2.234	2.254	2.274	2.274	2.305	2.350	2.398	2.446	2.449		
		Ebn at Noon	763	816	843	844	850	845	840	828	813	770	727	717		
		Edh at Noon	66	87	108	129	131	130	128	119	104	85	66	59		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.01	1.79	3.03	4.32	5.06	5.92	5.58	4.85	3.68	2.26	1.16	0.81		
		RadStd	0.14	0.30	0.47	0.63	0.53	0.52	0.52	0.47	0.36	0.20	0.13	0.15		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (101 neighbors)	N/A	N/A	N/A	N/A	+0.38	+0.29	N/A	N/A	N/A	+16

Nomenclature: See separate page